A Sex-Specific Analysis of Osteoarthritic Progression Using In Vitro Simulated Microgravity Exposure

Developing Robust Quantification Methodology for OA Progression and Sex Differences

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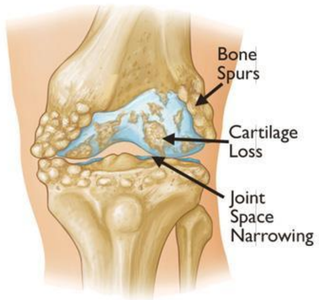
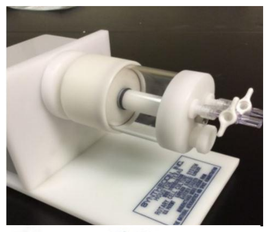
Sex Matters in Knee Osteoarthritis (OA)

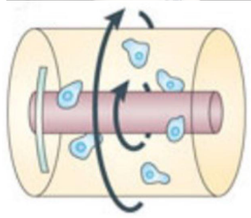


- Degeneration of the knee joint
- Affects all structures in the joint
- Mechanism not fully understood
- Severity not completely quantified

Knee osteoarthritis (KOA) incident rate and severity is disproportionately higher in females[2]

SMG-RVW Bioreactor

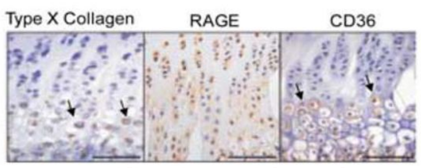




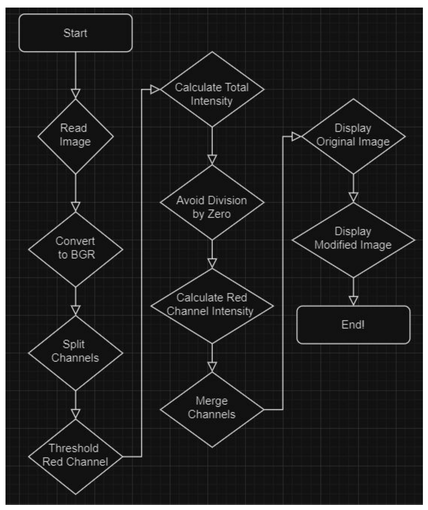
Rotating Wall Vessel (RWV) Bioreactor

- Horizontally rotated cylinder with a coaxial oxygenator.
- The vessel completely filled with media rotates as a **solid body.**
- **Induces MG** on the meniscus samples as there is no relative motion of the samples to the media.
- **Novel platform** for researching markers and replicating space conditions

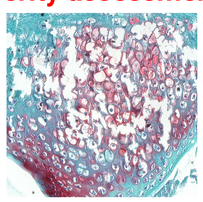
Future Translational Research



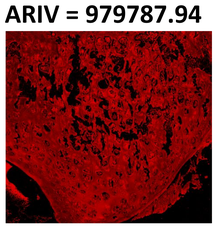
Quantification with Python



- Gender-specific responses to CD36 marker reveals heightened OA vulnerability in females.[1]
- Utilizing red intensity quantification in biomarker-stained tissues **enables OA severity assessment.**



ARIV = 979787.94

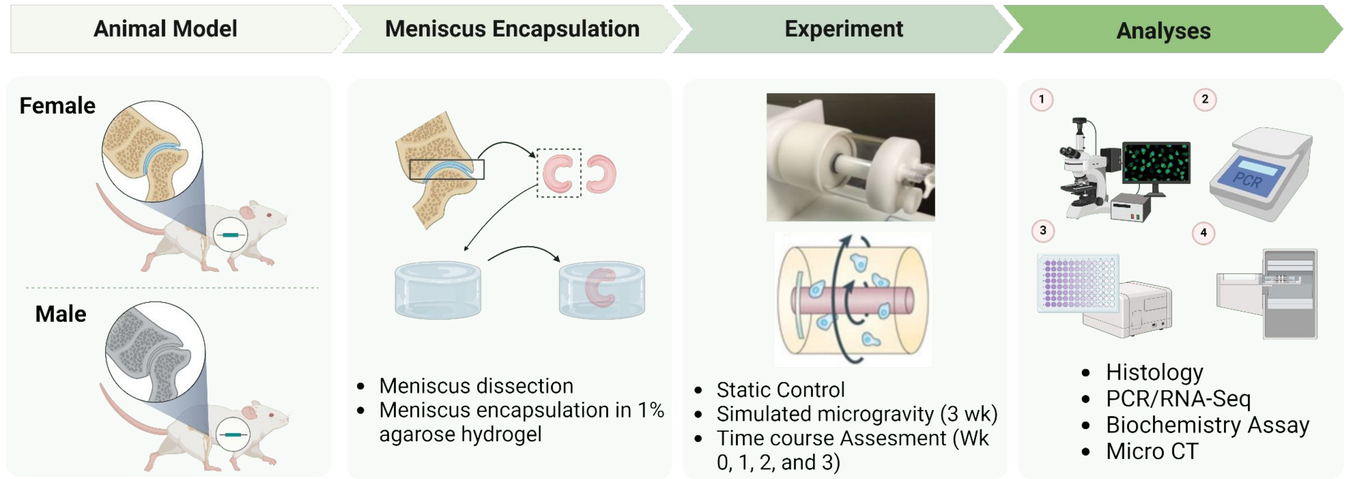


Specific Conclusion

- This sample does not exhibit signs of an OA phenotype due to the relatively high ARIV, as the GAG content is high.
- Correlation of this ARIV value with CD36 marker yields promising preliminary results.

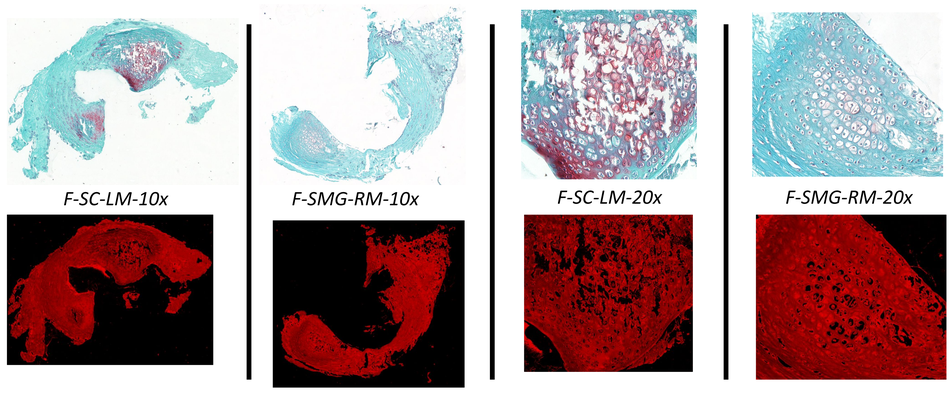
Methods

Hypothesis: Long term exposure to SMG will induce OA-like phenotype in meniscus excised from C57BL/6 mice. This platform will be used to study the sex differences, which can be quantified.

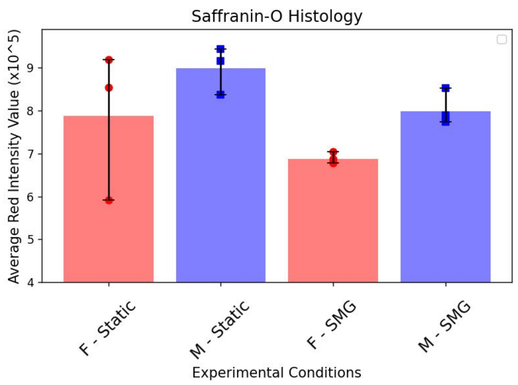


Results

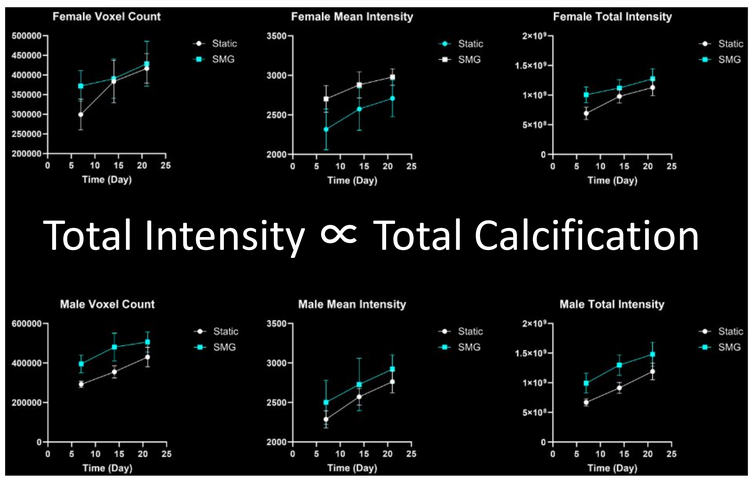
Histological Analysis - Safranin-O Staining - GAG/DNA



Quantitative Histology Results



CT Scan Analysis - Micro CT (uCT) - Calcification



- Safranin-O staining confirms literature findings.
- Simulated microgravity worsens OA phenotypes in both sexes.
- Females exhibit more severe OA phenotypes than males, **exacerbated by SMG**
- Python script **accurately quantifies OA severity**.
- SMG platform **offers method for finding OA markers.**
- 2D quantification correlates with 3D calcification assessment.
- Methodology is **robust, efficient, and novel**.
- Efficient platform to create OA phenotype

Conclusion

- SMG is a good platform for inducing OA, whereby sex-specific responses to SMG reveal **more aggressive OA phenotype in females.**
- Utilizing **red intensity quantification** in tissues with biomarkers enables **OA severity assessment between sexes, revolutionizing personalized healthcare.**

Funding

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References:

1. Cecil, D. L., Appleton, T. G., Polewski, M. D., Mort, J. S., Ann Marie Schmidt, Bendele, A., Beier, F., & Terkeltaub, R. (2009). The Pattern Recognition Receptor CD36 Is a Chondrocyte Hypertrophy Marker Associated with Suppression of Catabolic Responses and Promotion of Repair Responses to Inflammatory Stimuli. Journal of Immunology, 182(8), 5024–5031. https://doi.org/10.4049/jimmunol.0803603
2. Ma, Z., Li, D. X., Chee, R. K. W., Kunze, M., Mulet-Sierra, A., Sommerfeldt, M., Westover, L., Graf, D., & Adesida, A. B. (2022). Mechanical Unloading of Engineered Human Meniscus Models Under Simulated Microgravity: A Transcriptomic Study. Scientific Data, 9(1). https://doi.org/10.1038/s41597-022-01837-x